Xiang Zhang

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Graduate Studies

Princeton University *2021-present*
Ph.D. Candidate in Economics
Expected Completion Date: May 2027

References

Professor Gene Grossman
Department of Economics
Princeton University
609-258-4032
grossman@princeton.edu

Professor Eduardo Morales
Department of Economics
Princeton University
609-937-5412
ecmorale@princeton.edu

Professor Stephen Redding
Department of Economics
Stanford University
reddings@stanford.edu

Professor Richard Rogerson
Department of Economics
Princeton University
609-258-4839
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Prior Education

University of Chicago *2017 - 2019*
M.A. in Computational Social Science (Econ track)

Renmin University of China *2012 - 2016*
B.S. in Mathematics and Applied Mathematics

Peking University *2014 - 2016*
B.A. in Economics

Fields

International Trade, Macroeconomics, Spatial Economics, Urban Economics

Job Market Paper

Scale without Scope: Firm Specialization and Rising Industry Concentration (*JMP*)

Abstract: U.S. industries have become increasingly concentrated, with large firms accounting for a growing share of economic activity. Using U.S. manufacturing firm-product data from 1990 to 2020, I show that rising concentration is accompanied by increasing specialization among industry leaders. The average number of products produced by the top 1% of firms declines by more than 40%. I develop and estimate an oligopolistic model of multi-product firms, in which firms choose the product scope and invest in scalable, product-specific technology investments. Better scalable technology encourages firms to invest in their strongest products, while competition and higher internal investment costs make peripheral products less attractive and reduce their product scope. Firms can therefore grow larger while producing fewer products. I quantify the model through an application to information technology (IT), an example of scalable production investment. Improvements in IT efficiency between 1990 and 2020 account for nearly one-third of the observed rise in industry concentration and generate narrower product portfolios.

Working Papers

1. "The Economics of Agglomeration: Evidence from Grocery Stores" (with August Chen, Franklin Qian, and Qianyang Zhang).
   Revise and Resubmit at *Management Science*.

   Abstract: We provide the first large-scale causal estimates of demand-side agglomeration spillovers in the non-tradable service sector, leveraging data on 851 grocery store openings across the United States between 2019 and 2022. We combine a convolutional neural network with propensity score matching to identify credible counterfactual opening sites and compare business outcomes between actual and counterfactual locations. Grocery store openings increase foot traffic to nearby businesses within 0.1 miles by 23.4 percent within 6–12 months, with spillovers decaying sharply beyond this radius. The effects are strongest for wholesale and retail establishments and hospitality services, and are substantially attenuated after COVID-19—from 42.7 to 14.6 percent—consistent with a trip-chaining mechanism. Grocery anchors also reshape the local business landscape: within 0.1 miles, establishment growth rates rise by 7.0 percentage points and employment growth rates by 8.8 percentage points, averaged over the first three post-entry years. Decomposing establishment growth rates, the effect is accounted for by higher entry and fewer exits.

2. "Occupation Dynamics in General Equilibrium" (with Benny Kleinman, Ernest Liu, and Stephen Redding).

   Abstract: We develop a tractable quantitative model of automation and non-neutral technical change with dynamic labor mobility and capital accumulation. Given data on an economy somewhere along the transition path to an unobserved steady-state, we show how to solve for this unobserved steady-state and the counterfactual transition path in response to shocks. We invert the equilibrium conditions of the model to recover shocks that exactly rationalize the observed employment and wage data. We find relatively rapid convergence to steady-state, such that initial conditions are relatively unimportant in explaining the observed employment and wage dynamics. We find that intermediate-wage occupations have experienced the greatest automation, whereas high-wage occupations have witnessed the greatest productivity growth. We find that the combination of automation and productivity growth explains most of the observed employment and wage polarization in the data. We compute an inequality-output frontier, which summarizes the impact of automation and productivity growth on wage inequality and aggregate output, and can be used to assess the effects of emergent new technologies, such as artificial intelligence.

3. "Firm-to-firm relationship, uncertainty, and trade"

   Abstract: The recent surge in trade and industry policy uncertainty has led to growing disruptions in global value chain relationships, posing additional risks for exporters. I study the effects of relationship termination risk on exporter performance, investment, and search decisions. Event-study analysis highlights that the loss of major international buyers has a long-lasting detrimental impact on exporter performance. Firms losing major buyers experience a decline in export value and an increased likelihood of exiting the export market. These firms intensify their search efforts after the loss, yet their newly formed relationships tend to be short-lived. Inspired by these facts, I build an export dynamics model with search frictions and endogenous relationship-specific investment. Calibration results indicate that exporting firms face substantial search costs, as suggested by the existing literature. Moreover, relationship termination risk, and its subsequent influence on firms' endogenous choices, are primary factors explaining the observed firm dynamics.

4. "On the role of health in climbing the income ladder: Evidence from China" (with Gordon Liu and Franklin Qian).

Work in Progress

5. "Input Specificity, Capacity Constraint, and Supply Chain Resilience"

6. "The labor market dynamics of U.S. inventors" (with Meghana Gaur).

Conferences and Seminars

2025	NBER SI Micro and Macro Perspectives of the Aggregate Labor Market
2024	UEA European Meeting

Research Experience

2022-2023	Research Assistant for Professor Stephen Redding, Princeton University
2021-2022	Research Assistant for Professors Gregor Jarosch and Ezra Oberfield, Princeton University
2019-2021	Senior Research Specialist, International Economics Section, Princeton University. Assisted with research projects for Professors Gene Grossman, Eduardo Morales, and Esteban Rossi-Hansberg

Teaching

Princeton	Macroeconomics - Fall 2026
	Introduction to Macroeconomics - Spring 2026
	Food and Agricultural Economics - Spring 2024, Spring 2025
Chicago Booth	Microeconomics - Fall 2018

Honors Awards Scholarships and Fellowships

Avinash K. Dixit Prize, International Economics Section, Princeton University *For Outstanding Achievements in International Economics*	*2024*
International Economics Section Summer Fellowship, Princeton University	*2023 - 2025*
Real Estate Research Institute (RERI) funding	*2022*
Graduate Fellowship, Princeton University	*2021 - 2027*
Social Science Scholarship, The University of Chicago	*2017 - 2019*

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